

RideWire Job Aid — P0371 / P0372 / P0374

Per Harley-Davidson Service Bulletin M-1140 ("EDM Update", issued October 8, 2003), the correct DTCs for the crank position sensor on carbureted models are P0371 (Crankshaft position shorted low / CKP sensor wrong number of pulses), P0372 (Crankshaft position shorted high), and P0374 (Crankshaft position sensor not detected / cannot synchronize).

*This job aid is AI-assisted research support compiled from real Harley-Davidson service bulletins, diagnostic manuals, and owner/technician forum reports, cross-checked by three independent AI models. It is **not** a certified repair procedure and does not replace the vehicle's factory service manual. Always verify against manufacturer documentation before performing repairs.*

Platform

Harley-Davidson carbureted models (2004 model year Softail, Touring, Dyna, Sportster) - Crankshaft Position (CKP) sensor circuit, per official Harley-Davidson Technical Service Bulletin M-1140

Code Description¹

Per Harley-Davidson Service Bulletin M-1140 ("EDM Update", issued October 8, 2003), the correct DTCs for the crank position sensor on carbureted models are P0371 (Crankshaft position shorted low / CKP sensor wrong number of pulses), P0372 (Crankshaft position shorted high), and P0374 (Crankshaft position sensor not detected / cannot synchronize). The bulletin explicitly corrected earlier 2004 diagnostic manuals that had incorrectly listed P0373 and P0374 for this fault; P0373 is stated as "not available." A DTC in this family sets if the CKP signal is weak, shorted, or absent.

Likely Causes (verified sources)

- Intermittent connection, pinched or damaged CKP sensor wiring, or a loose CKP sensor mounting fastener
- Failed or degraded crankshaft position sensor unable to produce the minimum required AC signal (per the HD bulletin's diagnostic chart, at least 1 VAC during cranking) at its output terminals
- Wiring shorted to ground or to voltage between the CKP sensor connector and the ignition control module (ICM)

Recommended First Check²

Per the official Harley-Davidson diagnostic flowchart in TSB M-1140, first check for intermittent connections, pinched/damaged wires, and a loose CKP sensor fastener; if none are found, connect a breakout box to the harness (leaving the ignition control module disconnected) and measure resistance between the CKP sensor terminals and ground -- resistance greater than 1 megohm is expected for a good circuit, with further AC-voltage cranking tests (minimum 1 VAC) used to isolate a bad sensor from a wiring fault.

Source confidence: Strong: primary source is the actual scanned/hosted text of official Harley-Davidson Technical Service Bulletin M-1140 (an EDM/diagnostic-manual correction notice), independently corroborated by a separately-compiled dealer-sourced DTC reference document (hosted on S3, aggregating Harley diagnostic manuals) and further discussed by riders on an independent Harley owner forum reporting the same P0371/P0374 code pairing and description in the field.

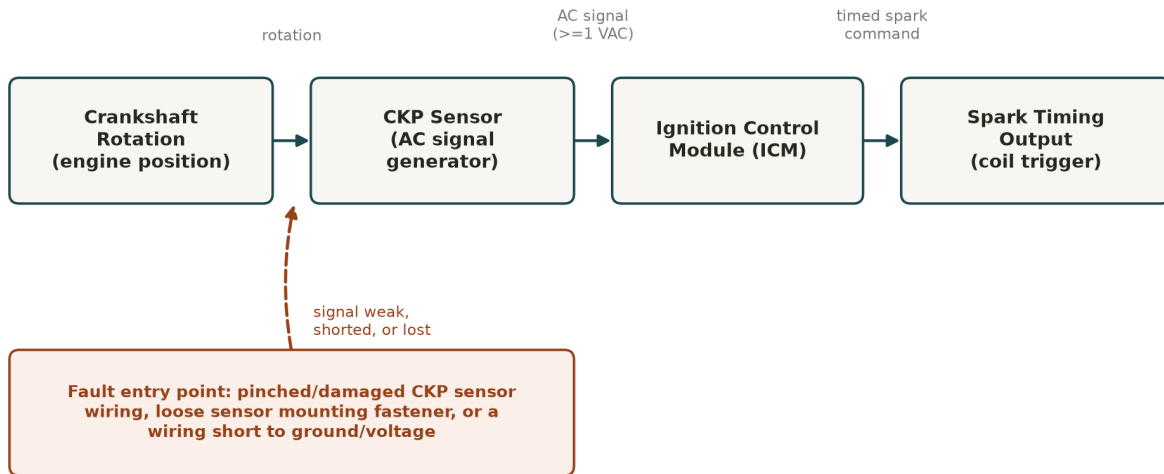
Reference Schematic

Generic principle-level diagram — not a manufacturer-exact wiring diagram for any specific make/model/year. Always verify against the factory service manual.

GENERIC REFERENCE: CKP Circuit (P0371 / P0372 / P0374)

Shows the PRINCIPLE of the crank position signal path - not a vehicle-specific wiring diagram

Result: ICM sees a weak, shorted, or absent CKP signal (not ≥ 1 VAC while cranking) -> cannot synchronize spark timing -> sets P0371/P0372 (signal shorted low/high) or P0374 (not detected)



Reference only - always verify against the actual vehicle's factory service manual before repair.

Multi-AI Consensus (3 independent models)

"Agreement" below is a plain keyword-overlap check on each model's own wording for this single query — it is NOT a validated machine-learning accuracy score and has not been checked against confirmed repair outcomes. Read each model's own words in the table for the actual reasoning.

Model	Primary Root Cause	Urgency	Recommended First Check
Claude Sonnet 5	Damaged, corroded, or high-resistance CKP sensor wiring/connector (open or shorted circuit between sensor and ECM)	high	Inspect the CKP sensor connector and wiring harness for corrosion, damage, or chafing, then back-probe the circuit with a multimeter/scope to verify sensor resistance and AC signal output match factory spec while cranking
GPT-5.6 Terra	Failed or intermittent crankshaft position sensor	high	Disconnect the CKP sensor connector and inspect the connector and harness closely for oil intrusion, chafing, broken conductors, or loose terminals, then test sensor resistance and continuity against the service-manual specification.
Gemini 3.1 Pro	Faulty Crankshaft Position (CKP) sensor	high	Remove the CKP sensor to inspect the magnetic tip for metallic debris and test sensor resistance/AC voltage output while cranking.

Root cause agreement: 3 of 3 models shared the keyword(s) ['sensor'] in their primary root cause

Urgency values reported: high, high, high (all 3 models agree)

Sources

1. <https://sportsterpedia.com/lib/exe/fetch.php/pdf-bulletin:tsb1140.pdf>
2. <https://s3.amazonaws.com/tnmarketing/fmh/Troubleshoot-with-Harley-Error-Codes.pdf>
3. <https://www.harley-davidsonforums.com/threads/crank-with-no-start-with-ckp-error-codes.378654/>

Superscript 1-2 in the text above refer to the same verified source set backing the code description and first-check procedure, drawn from the sources listed here.